

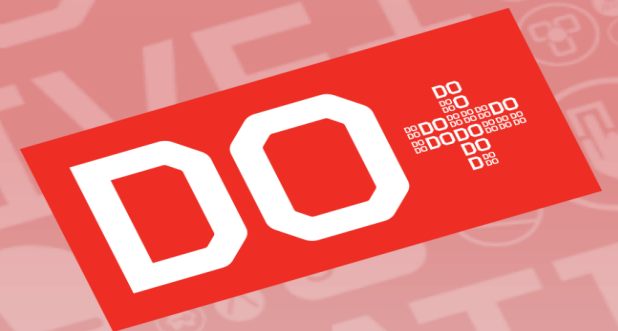


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XClarity™ Administrator

EBG Server Education Monitoring/Eventing Component

Andy Vrabel April 15, 2015



- Responsible for:
 - Logging hardware and audit events from managed systems
 - Recovering events lost due to communication errors, server down-time, etc
 - Logging internal event/audit conditions
 - Displaying/filtering active alert conditions
 - Determining overall endpoint status
 - Providing REST APIs for event/alert retrieval and management
 - Forwarding events to configured syslog, SNMP, and email managers/servers

Event Capture and Logging

- CMM, Blade and Rack server hardware event and audit events received through CIM indications
- Switch events received through SNMP traps and converted to Event class objects
- Historical events are not gathered
 - Events gathered only after event registration in place after managing
- Events deleted when an endpoint is unmanaged
 - Does not include internal audit events for the endpoint
- Internal system/audit events also raised
 - represent conditions/issues internal to the server
 - e.g. deploy success/failure, manage system success/failure
- Event text shipped as part of the LXCA image
 - Ships in English, simplified Chinese, and Japanese translations
 - Spanish scheduled for the June fixpack
- All events stored in the LXCA DB (LDAP) and restored on system start-up

Event Capture and Logging – Monitoring->Event Logs view

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StatusJobsUSERID

DashboardHardwareProvisioningMonitoringAdministration

Logs

Event LogAudit Log

The event log provides a history of hardware and management conditions that have been detected.



All Actions

Excluded Events

Show:   

All Event Sources

All Dates

Filter

☐	Severity	Serviceability	Date and Time	System	Event	System Type
☐	Informational	Not Required	Apr 6, 2015, 1:31:47 PM	Management Server	Call Home Event with ID 35010483 from component Chassis will not be called home because the server Call Home function is disabled.	Management
☐	Informational	Not Required	Apr 6, 2015, 1:31:47 PM	Management Server	Call Home Event with ID 35010483 from component Chassis has been added to the Call Home Processing queue.	Management
☐	Warning	Support	Apr 6, 2015, 1:31:46 PM	SN#Y030BG4BA020	Node Node 03 device Storage back plane[01] VPD is not valid.	Chassis
☐	Informational	Not Required	Apr 6, 2015, 1:30:33 PM	Kestrel_blue	Host Power has been turned on.	Server
☐	Informational	Not Required	Apr 6, 2015, 1:30:25 PM	Kestrel_blue	Attempting to Power On server SN# 06PCA68 by user SYSMGR_FTVRRYLC.	Server
☐	Informational	Not Required	Apr 6, 2015, 1:30:06 PM	Management Server	Minimum SSL/TLS protocol level on SN#Y030BG4BA020 matches minimum SSL/TLS protocol level legacy on the management server.	Management
☐	Informational	Not Required	Apr 6, 2015, 1:30:06 PM	Management Server	Cryptographic mode on SN#Y030BG4BA020 matches cryptographic mode nistcomp on the management server.	Management
☐	Warning	Not Required	Apr 6, 2015, 1:25:05 PM	Management Server	Cryptographic mode unknown on SN#Y030BG4BA020 does not match cryptographic mode nistcomp on the management server.	Management
☐	Warning	Not Required	Apr 6, 2015, 1:25:05 PM	Management Server	Minimum SSL/TLS protocol level unknown on SN#Y030BG4BA020 does not match minimum SSL/TLS protocol level legacy on the management server.	Management
☐	Informational	Not Required	Apr 6, 2015, 1:24:03 PM	Nantahala RED	Security: Userid: SYSMGR_FTVRRYLC had 1 login failures from a CIM client at IP address ::ffff:10.243.15.18.	Server
☐	Informational	Not Required	Apr 3, 2015, 1:26:56 PM	Management Server	Connectivity to chassis SN#Y030BG4BA020 has been restored. UUID is 63E29269BB634AB9A610D6F8FCE2B28F.	Management
☐	Warning	Not Required	Apr 3, 2015, 9:25:14 AM	Management Server	Minimum SSL/TLS protocol level unknown on SN#Y034BG17606E does not match minimum SSL/TLS protocol level legacy on the management server.	Management

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Event Capture and Logging – System->Event Logs view

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Actions

Flex System 81

Offline

General

- Summary
- Inventory Details

Status and Health

- Alerts
- Event Log**
- Jobs
- Light Path
- Power and Thermal

Configuration

- Feature on Demand Keys

Chassis > Flex System 81 > Flex System 81 - Event Log

Event LogAudit Log

? The event log provides a history of hardware and management conditions that have been detected.

 | All Actions |  |  | Show:    All Event Sources

☐	Severity	Serviceability	Date and Time	Event
☐	 Warning	 Support	Apr 8, 2015, 8:12:40 AM	Power supply Power Supply 03 power meter is offline.
☐	 Warning	 Support	Apr 7, 2015, 6:39:28 PM	Power supply Power Supply 03 power meter is offline.

Event Capture and Logging – Details and Recovery text

Severity: Informational

Attempting to Power On server SN# 06PCA68 by user SYSMGR_FTVRRYLC. 4000000F00000000

Properties	Details
Chassis:	SN#Y030BG4BA020
Bay:	Bay 3
System UUID:	687905B03D9311E186DB5CF3FC112EF2
System Serial Number:	06PCA68
System Type-Model:	7906-AC1
FRU Number:	90Y4734
FRU Serial Number:	Y234UF1BR05E
Event Source:	1

Apr 6, 2015, 1:30:25 PM

Severity: Informational

Attempting to Power On server SN# 06PCA68 by user SYSMGR_FTVRRYLC. 4000000F00000000

Properties	Details
Details: The specified user has initiated a power function on the system using the Management Controller.	
Recovery: Information only; no action is required.	

Apr 6, 2015, 1:30:25 PM

Active Alert Capture and Logging

- Represents current active alarm conditions
- Polled/refreshed as part of hardware inventory for external alerts
 - Polling triggered by receipt of a non-audit event or inventory refresh timer expiration
 - Correlated to LXCA event log for additional information about the condition
 - Can only be correlated if the alert was raised after the endpoint was managed since historical events are not collected
 - Condition clears when the endpoint no longer reports it
 - Not stored in the LXCA DB
- Internal alerts raised and cleared by plug-ins/internal components
 - Must be cleared by the component that raised it
 - Not persisted across an LXCA restart
- Overall health status provided by summarizing alerts raised against each endpoint
 - Health status may not match endpoint since internal alerts are also considered

Active Alert Capture and Logging - System->Alerts view

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StatusJobs

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Actions

SN#Y030BG4BA020
Critical

General

- Summary
- Inventory Details

Status and Health

- Alerts**
- Event Log
- Jobs
- Light Path
- Power and Thermal

Configuration

- Feature on Demand Keys

Chassis > SN#Y030BG4BA020 > SN#Y030BG4BA020 - Alerts

Alerts indicate hardware or management conditions that need investigation and user action.

 All Actions 

Show:    All Alert Sources All Dates

☐	Severity	Serviceability	Date and Time	Alert
☐	Warning	Not Available	Apr 6, 2015, 8:52:17 PM	Node Node 03 message: System firmware, firmware progress (GPT Status) warning.
☐	Warning	Support	Apr 6, 2015, 8:51:54 PM	Node Node 03 device Storage back plane[01] VPD is not valid.
☐	Critical	Not Available	Apr 6, 2015, 8:40:09 PM	Node Node 11 - 01 failed initial provisioning.
☐	Warning	Not Available	Apr 6, 2015, 8:37:07 PM	Hot air exiting from the rear of the chassis might be recirculated in the inlet air at the front of the chassis. Slot 12: 29.00C, Slot 1: 21.50C.
☐	Warning	Not Available	Apr 6, 2015, 8:21:28 PM	Chassis Management Module Standby CMM is not compatible. Redundant capability is turned off.
☐	Warning	Not Available	Apr 6, 2015, 8:20:11 PM	Fan logic module Fan Logic 02 is an older revision card (FRU 81Y2912) and needs to be replaced.
☐	Warning	Not Available	Apr 6, 2015, 8:20:10 PM	Fan logic module Fan Logic 01 is an older revision card (FRU 81Y2912) and needs to be replaced.
☐	Warning	Not Available	Apr 6, 2015, 8:19:51 PM	Cooling zone zone01 might not have adequate cooling.
☐	Critical	Not Available	Apr 6, 2015, 8:19:50 PM	Fan module Fan 01 has failed.
☐	Critical	Not Available	Apr 6, 2015, 8:19:46 PM	Fan module Fan 07 has failed.
☐	Critical	Not Available	Apr 6, 2015, 8:19:45 PM	Fan module Fan 02 has failed.
☐	Warning	Not Available	Apr 6, 2015, 8:18:47 PM	The CMM J40 jumper is installed in bay 1.

Event/Alert Exclusion

- Events and alerts can be excluded from view and forwarding if they are of no interest to the customer
- Events/alerts excluded by one user are excluded for all users
 - Separate configuration of filters for events and alerts
- Scope is by system or all systems

Exclude Events

Select an Exclusion Rule to save. Exclusion rules and excluded events can be viewed by selecting the Excluded Events toolbar button.

- ☒ Exclude selected events from all systems.
- ☐ Exclude events only from systems in the scope of the instance selected.

Save

Close

Excluded Events

Exclusion Rules

Excluded Events

View events that have been excluded from the event log.



All Actions

Filter

Date and Time	System	Event ID	Event
2015-04-02T18:58:38Z	Management Server	FQXHMSE0006G	Cryptographic mode unknown on SN#Y034BG17606E does not match
2015-04-02T19:23:41Z	Management Server	FQXHMSE0006G	Cryptographic mode unknown on DaAn5 does not match cryptographic
2015-04-03T13:25:14Z	Management Server	FQXHMSE0006G	Cryptographic mode unknown on SN#Y034BG17606E does not match
2015-04-08T17:25:05Z	Management Server	FQXHMSE0006G	Cryptographic mode unknown on SN#Y030BG4BA020 does not match

Reliable Events/Event Recovery

- Responsible for recovering hardware and audit events
 - e.g. events lost due to communication/network issues or while the LXCA server is down
 - All CMM/IMM events are recoverable but for switches, only SAL (switch audit log) events are recoverable
- Uses the event sequence number to detect missed events
 - Endpoints can maintain multiple logs each with a separate sequence counter
- Order of event generation preserved
 - Events are queued internally while recovery is in progress
- Gaps in the record log may still exist if events are cleared from the endpoint or the CMM/IMM detects event corruption and re-sequences
- Does not apply to internal events

- Provides interfaces for
 - GET – get event log and active alert entries, active status, and event log configuration
 - Various filters supported for narrowing scope of retrieval
 - PUT – add internal event to the event log, change event log configuration, edit event monitor
 - 2000 system events + 2000 audit events stored before logs wrap
 - API available to change the defaults but UI does not currently exploit it
 - POST – internal active alert, event action, event action aggregation point, create event monitor
 - DELETE – active alert, event action, event monitor

Event Forwarding

- Very limited space available to store events so event forwarding recommended for users who want complete historical log record
- Remote syslog, SNMP trap, and email monitors supported
- Monitors can be disabled and re-enabled after configuring
- Filters available to monitor individual systems or all systems for selected event types/severities
 - Audit events are an all-or-none selection
- A total of 12 monitors can be configured
 - Maximum 2 can be syslog

New Event Recipient

General

Systems

Events

Select an event recipient type:

☐ Syslog

☐ SNMPv3

☒ Email

* Name

[?](#)

* Host

[?](#)

* Port

Description

* Recipients

[?](#)

Status

☒ Enable this recipient

- Syslog message format is structured to enable parsing
 - RFCs 3164 and 5424 used as guidelines but not as an architecture
 - Integration with VMware Log Insight key to design

<86> Mon Apr 06 21:29:53 EDT 2015 [appl=LXCA service=NONE severity=INFORMATIONAL class=AUDIT
appladdr=10.243.15.18 src=CMM uuid=63E29269BB634AB9A610D6F8FCE2B28F sn=23EWD29 seq=1692 EventID=00200000]
Remote login failed for user SYSMGR_QXVFARSJ from CIM at IP address 10.243.6.93

- Most fields are tagged to enable parsing
- English text only

- Only SNMPv3 traps supported
- Traps can optionally include authentication and privacy protocols
- LXCA supports a single, converged MIB for all traps
 - No need to gather MIBs from the supported CMMs, IMMs, switches. All forwarded events are mapped to traps within the LXCA MIB.
 - Internal events included
 - Published with the online documentation
- RFC 2574 USM for SNMPv3 support provided for user authentication verification
 - The USM MIB is the only one supported for SNMP GETs

New Event Recipient

General

Systems

Events

Select an event recipient type:

☐ Syslog ☒ SNMPv3 ☐ Email

* Name

Description

Contact Name

User ID

Authentication Type

None

Privacy Password

Status

☒ Enable this recipient

* Host

* Port

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Location

Authentication Password

Privacy Type

None

- Similar to Syslog monitor in that tagged fields are used to label event data
- Fields/subject line are not configurable
- E-mail encryption is not supported

From: LXCA@10.243.15.18

Sent: Thursday, April 02, 2015 3:13 PM

To: user@fvta6.labs.lenovo.com

Subject: Device Node in bay 1 was powered on by user ID USERID from Web at IP address 10.41.46.75

Alert: Device Node in bay 1 was powered on by user ID USERID from Web at IP address 10.41.46.75

Managed Endpoint: IBM:CMM01:Y034BG16F01F:00

Event ID Logged is: 40000022

LXCA is: 10.41.46.75

Event information:

Severity : INFORMATIONAL

Event Class : AUDIT

Sequence ID : 120

Source ID : 4c64e79d47e345dcb2f75c548a15ddc1

Serial Num : 23PYP15

MTM : 789392X

Service : None

- REST APIs are your best friend when troubleshooting eventing issues
- Chassis and Primary are treated the same wrt alerts and overall health status
- Some events discarded if they are assumed to be duplicates
 - EventIDs 7777 770x, x=1, 2, 3
 - CMM events for p-blades
 - Server events from secondary nodes in a scalable, partitioned configuration
- Debugging alert issues from the logs:
 - AssertedEventUtils – dumps alerts per endpoint upon retrieval
 - ActiveAlertServlet – dumps list of all alerts every 45 seconds (UI poll) plus filtered count if filtered by component ID
- Debugging event issues from the logs:
 - CIMIndicationListener – dumps contents of all events received
 - SnmpTrapListener – dumps contents off all switch traps received
 - ReliableEventService – enforces correct event sequencing
 - EventManager – Discards duplicate events
 - Events – dumps contents off all events before storing in database

THANK YOU GRACIAS
MERCI DANKE GRAZIAS 謝謝 СПАСИБО
OBRIGADO ありがとう DANK TAKK BEDANKT DAKUJEM